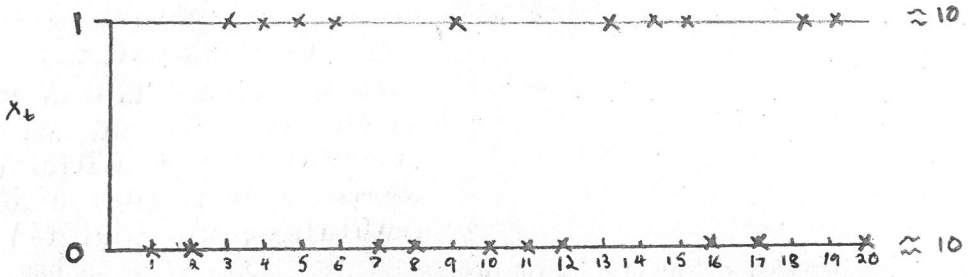
MA0367 Time Series Analysis and Forecasting homework 1 2016-17

Submission deadline: Friday, February 03

(1) Sketch and describe a realization of the stochastic process $x_1, x_2, \ldots, x_{20}$, where x_k has the Bernoulli distribution: 0 with probability 0.5 and 1 with probability 0.5, x_i and x_j are independent for $i \neq j$. Find the mean function and the variance function.

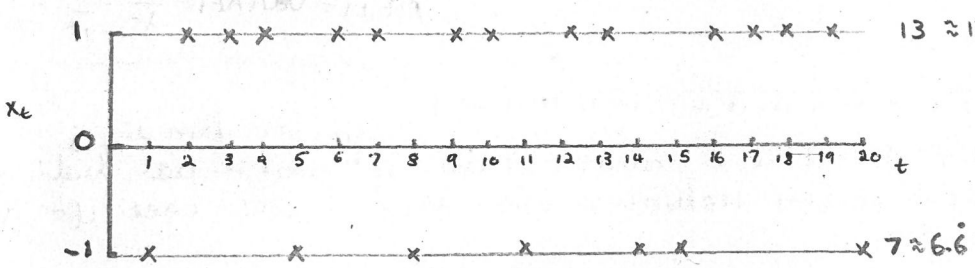


$M(t) = E(X_t) = 1(0.5) + 0(0.5)$
$\overset{\text{i.i.d}}{=} 0.5$

$R(t,t) = \text{Var}(X_t) = E(X_t^2) - E(X_t)^2$
$= 1^2(0.5) - 0.5^2$
$= 0.25$

— Each x_n is equally likely to take a value of 0 or 1. Thus after 20 observations (time increments), we would expect 10 realisations (values) on each line. The distribution of points on the line is randomly chosen.

(2) Sketch and describe a realization of the stochastic process $x_1, x_2, \ldots, x_{20}$, where x_k has the Bernoulli distribution: -1 with probability 1/3 and 1 with probability 2/3, x_i and x_j are independent for $i \neq j$. Find the mean function and the variance function.

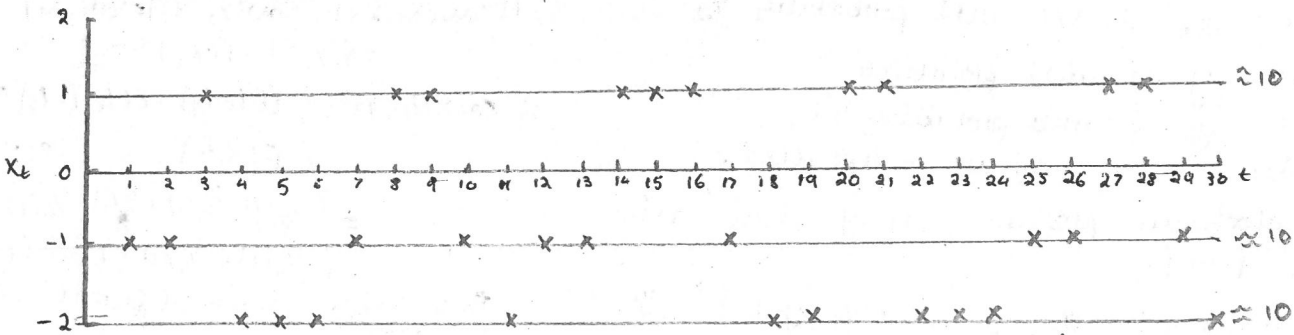


$M(t) = E(X_t) = 1(2/3) - 1(1/3)$
$\overset{\text{i.i.d}}{=} 1/3$

$R(t,t) = \text{Var}(X_t) = E(X_t^2) - E(X_t)^2$
$= \left(1^2(2/3) + 1^2(1/3)\right) - 1/9$
$= 8/9$

— Each x_n has a prob of $2/3$ to equal 1, and $1/3$ = prob of taking value of -1. Over 20 trials we would expect 13.3 points on 1 and 6.6 on -1. Since we cannot have decimals, the most likely case is 13 on 1 and 7 on -1.

(3) Sketch and describe a realization of the stochastic process $x_1, x_2, \ldots, x_{30}$, where x_k has the discrete distribution: -2 with probability 1/3, -1 with probability 1/3 and 1 with probability 1/3, x_i and x_j are independent for $i \neq j$. Find the mean function and the variance function.



$M(t) = E(X_t) = -2(1/3) - 1(1/3) + 1(1/3) = -2/3$
$\overset{\text{i.i.d}}{}$

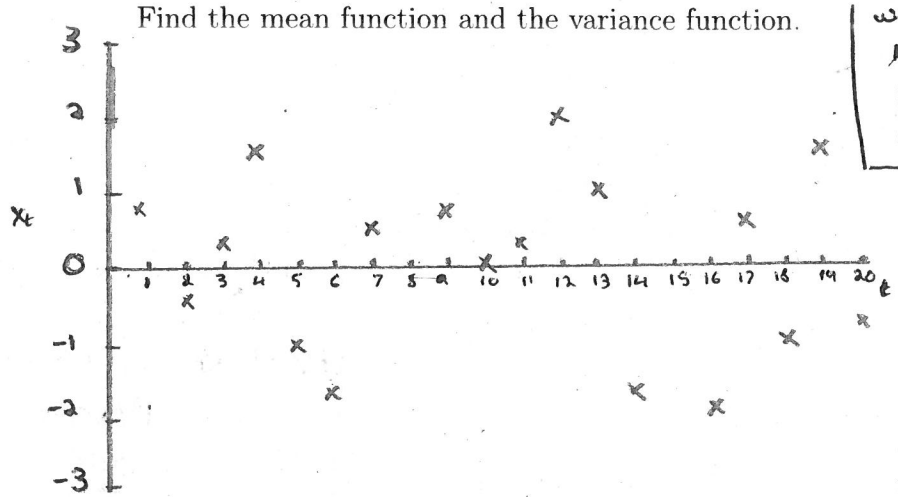
$R(t,t) = \text{Var}(X_t) = \left(4(1/3) + 2(1/3)\right) - 4/9 = \frac{14}{9}$

This is similar to previous examples with the additional feature x_n can take 3 values. We would expect 10 points on each line. How the points distribute on the lines have been randomly chosen

Note that this is one possible realisation.

(4) Sketch and describe a realization of the stochastic process $x_1, x_2, \ldots, x_{20}$, where x_k has the normal distribution with mean 0 and variance 1, x_i and x_j are independent for $i \neq j$. Find the mean function and the variance function.

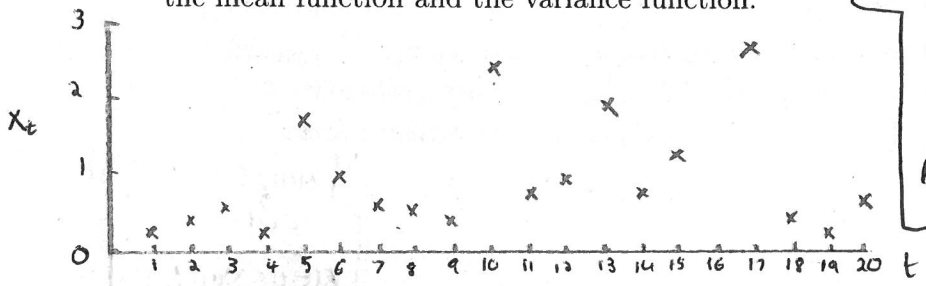
we have
$X_i \sim N(0,1)$ and i.i.d (independent)
$M(t) = E(X_t) = 0$
$R(t,t) = Var(X_t) = 1$



To obtain the possible values of x_k, we want to take a random sample from the standard normal distⁿ. This can be easily done in R. Note this is only one possible realisation and another random sample will produce a different realisation (very likely to).

(5) Sketch and describe a realization of the stochastic process $x_1, x_2, \ldots, x_{20}$, where x_k has the exponential distribution with parameter 1, x_i and x_j are independent for $i \neq j$. Find the mean function and the variance function.

we have
$X_i \sim \text{exp}(\lambda = 1)$ (negative exp distⁿ) and independent
$E(X_t) = M(t) = \frac{1}{\lambda} = \frac{1}{1} = 1$
$R(t,t) = Var(X_t) = \frac{1}{\lambda^2} = \frac{1}{1^2} = 1$



Like 4, we use R to take a random sample of size 20 from the exponential distⁿ where $\lambda = 1$. This is only one possible realisation and this is the case for all previous question too.

(6) Sketch and describe realizations of <u>two</u> discrete stochastic processes x_t which have the